Weekly Research Journal

Motor-Based Vehicle Velocity Estimation Using VESC and ROS 2

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F1TENTH Autonomous Vehicle Project

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Abstract

This week established a complete VESC-to-ROS 2 path for monitoring motor-based vehicle velocity. The VESC controller was identified on /dev/ttyACM1, connected with firmware version 6.6, and published /sensors/core at approximately 50 Hz. Motor ERPM was converted to /odom longitudinal velocity using the provisional gain 4614.0. A 0.05 m/s deadband initially suppressed all manual wheel-rotation samples; reducing it to 0.001 m/s allowed non-zero low-speed output and real-time rqt_plot visualization. The result verifies data transport and low-speed observability, but it is not an absolute vehicle-speed calibration.

1 Weekly contribution

The work combined hardware identification, ROS 2 interface verification, source-level diagnosis, and a controlled low-speed test. Table 1 states the demonstrated outcome and the limitation that remains.

Table 1: Evidence and scope of the 8 October results.

Contribution	Evidence	Scope
VESC telemetry	Firmware 6.6; /sensors/core at approximately 50 Hz	Completed
ROS 2 velocity path	ERPM converted to /odom and displayed in rqt_plot	Completed
Low-speed visibility	Deadband changed from 0.05 to 0.001 m/s	Functional test
Absolute speed	No measured-distance reference; gain 4614.0 provisional	Not calibrated

2 System configuration and data path

USB enumeration showed the Hokuyo LiDAR at /dev/ttyACM0 and the STMicroelectronics VESC interface at /dev/ttyACM1. Using stable device identity was important because the numeric ttyACM assignment changed after power cycling. Once the VESC driver used the correct port, the controller reported firmware version 6.6 and entered its operating state.

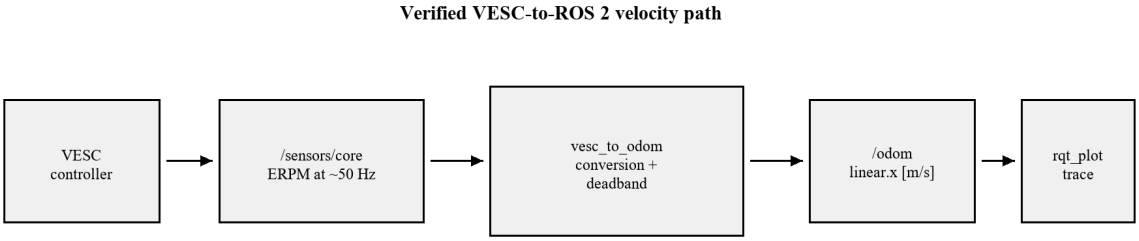


Figure 1: Verified motor-speed data path from the VESC controller to the ROS 2 odometry topic and real-time plotting interface.

Table 2: Verified interfaces and parameters.

Item	Observed value	Interpretation
VESC USB device	/dev/ttyACM1	Correct controller port after restart
Firmware	6.6	Driver initialization completed
Telemetry rate	approximately 50 Hz	Stable /sensors/core publication
Conversion gain	4614.0 ERPM per m/s	Provisional, not calibrated
Final test deadband	0.001 m/s	Allows manual low-speed observation

2.1 Motor-based velocity conversion

The odometry node converts electrical motor speed to a signed longitudinal estimate. With zero offset, the implemented relation is

$$v_{VESC} = -ERPM / 4614.0. \quad (1)$$

This is an encoder-based motion proxy. It can respond quickly, but it does not directly measure the vehicle's displacement relative to the ground and can therefore differ from actual speed during wheel slip, unloaded wheel rotation, or external pushing.

3 Low-speed diagnosis and validation

3.1 Deadband failure mechanism

Manual rotation produced reported ERPM magnitudes up to 69. Equation (1) maps 69 ERPM to approximately 0.0150 m/s. The original 0.05 m/s deadband corresponds to about 231 ERPM, so every manual sample was deliberately forced to zero. The absence of /odom motion was therefore a filter effect rather than a serial-communication failure.

$$\mathrm{ERPM}_{\mathrm{threshold}} = 0.05 \times 4614 = 230.7 \text{ ERPM}. \quad (2)$$

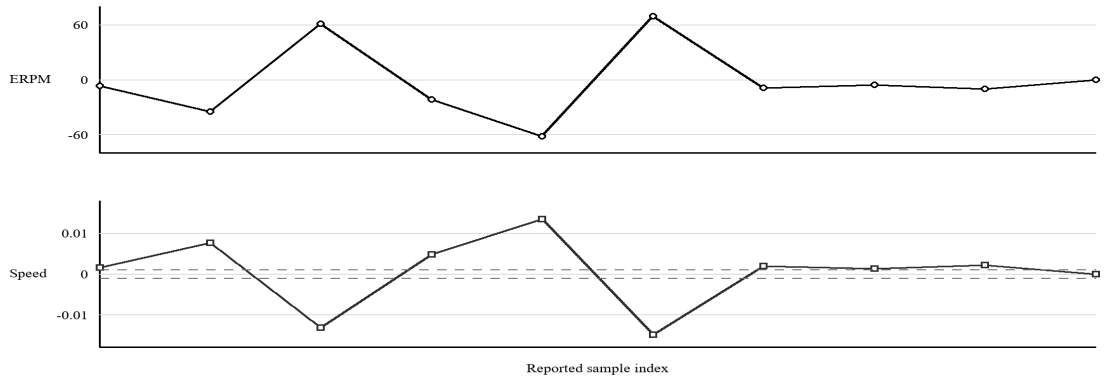


Figure 2: Reported manual wheel-rotation ERPM samples and their provisional velocity conversion. The dashed lines mark the final +/-0.001 m/s test deadband. Sample index is used because a synchronized time record was not retained.

Table 3: Selected reported ERPM values and provisional conversion.

ERPM	Converted speed [m/s]	Original 0.05 m/s deadband
-35	+0.0076	Suppressed
61	-0.0132	Suppressed
-62	+0.0134	Suppressed
69	-0.0150	Suppressed

3.2 Observed result

After changing the deadband to 0.001 m/s and rebuilding vesc_ackermann, /odom displayed non-zero values during wheel rotation and returned to zero when rotation stopped. The same signal appeared in rqt_plot. This validates the software path and its real-time response at low speed. It does not validate the physical scale of Equation (1).

4 Completed work and open work

Table 4: Project status at the end of the reporting week.

Status	Item
Completed	Identified Hokuyo and VESC serial devices after power cycling.
Completed	Verified VESC firmware 6.6 and approximately 50 Hz telemetry.
Completed	Published motor-based longitudinal velocity on /odom.
Completed	Resolved low-speed suppression by changing the deadband to 0.001 m/s.
Completed	Displayed the velocity signal in real time with rqt_plot.
Pending	Calibrate speed_to_erpm_gain against measured distance and time.
Pending	Quantify bias, repeatability, delay, and sign convention.
Pending	Compare motor-based speed with whole-vehicle Cartographer motion.

5 Discussion and next experiment

The central achievement is a reliable path from VESC motor telemetry to a visible ROS 2 velocity estimate. The deadband analysis also demonstrates why a valid sensor stream can produce an apparently constant zero at the application level. The remaining limitation is measurement validity: motor ERPM is not an independent observation of whole-vehicle ground speed, and the conversion gain is still a nominal value.

The next stage should obtain Cartographer position and yaw, differentiate position to estimate map-frame vx, vy, speed, and movement direction, and compare that result with VESC odometry. A measured-distance run is required before reporting speed error or treating 4614.0 as calibrated.

Planned validation sequence

Step	Measurement	Decision enabled
1	Measured distance and elapsed time	Calibrate ERPM-to-speed scale
2	Repeated forward and reverse runs	Check sign and repeatability
3	Cartographer pose-derived velocity	Compare motor and body motion
4	Synchronized residuals	Assess later sensor fusion